

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch : Artificial Intelligence & Allied Semester :III

Subject Code & Name: BTAIC303 & Data Structure and Algorithm using Python

Max Marks: 60

Date:06-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Write a program to reverse the given string	(BT3/CO1)	6
B) What is mean by operators? List all operators and explain any two.	(BT2/CO1)	6
C) Explain the use of break and continue statement with suitable example	(BT2/CO1)	6
Q.2 Solve Any Two of the following.		12
A) What are functions? Explain different functions types with example.	(BT2/CO2)	6
B) What is class and object? Write a program to initialize and display details of two student using classes and objects?	(BT3/CO2)	6
C) What is Exception Handling? Explain how to handle exceptions?	(BT3/CO2)	6
Q.3 Solve Any Two of the following.		12
A) Write algorithms for detecting node from Doubly linked list	(BT3/CO3)	6
B) What is mean by array? Explain with suitable example	(BT2/CO3)	6
C) Explain the concept of Stack with its operations.	(BT2/CO3)	6
Q.4 Solve Any Two of the following.		12
A) Write python program for Binary Search algorithm.	(BT3/CO4)	6
B) Define tree data structure? Explain Left and Right Rotation.	(BT2/CO4)	6
C) Define hashing Techniques with its factions and application	(BT2/CO4)	6
Q.5 Solve Any Two of the following.		12
A) Explain Insertion and Quick sort with algorithm.	(BT3/CO5)	6
B) What is an algorithm? Explain time complexity and space complexity	(BT2/CO5)	6
C) Write algorithm for insertion sort? Sort following element using insertion sort 40,30,20,50,45,60,10.	(BT3/CO5)	6

*** End ***